United States Patent Application Publication 20250279234
Kind Code A1
Publication Date September 04, 2025
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WOUND CORE

Abstract

In this wound core, the average distance of first-group joint portions and the average distance of second-group joint portions determined under predetermined conditions satisfy predetermined conditions.

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Appl. No.:	**18/857522**
Filed (or PCT Filed):	**June 21, 2023**
PCT No.:	**PCT/JP2023/022934**

Foreign Application Priority Data

JP	2022-100293	Jun. 22, 2022

Publication Classification

Int. Cl.:	**H01F27/245** (20060101)
U.S. Cl.:	
CPC	**H01F27/2455** (20130101);

Background/Summary

TECHNICAL FIELD OF THE INVENTION

[0001] The present disclosure relates to a wound core.

[0002] The present application claims priority based on Japanese Patent Application No. 2022-100293 filed in Japan on Jun. 22, 2022, the contents of which are incorporated herein by reference.

RELATED ART

[0003] A wound core is widely used as a magnetic core for a transformer, a reactor, a noise filter, or the like. Conventionally, reduction of iron loss occurring in a core has been one of important problems from the viewpoint of high efficiency and the like, and reduction of iron loss has been studied from various viewpoints.

[0004] For example, Patent Document 1 discloses a wound core in which a plurality of core materials each having at least one cutting portion are wound for each winding and a rectangular window portion is provided at the center, in which a space factor of the core material at a corner portion is lower than a space factor of the core material at a side portion excluding the corner portion.

CITATION LIST

Patent Document

[0005] Patent Document 1: Japanese Unexamined Patent Application, First Publication No. 2015-141930

SUMMARY OF INVENTION

Problems to be Solved by the Invention

[0006] Currently, there is a demand for a wound core in which noise is suppressed more than in the case of Patent Document 1.

[0007] The present disclosure is an invention that has been made in view of the above problems, and provides a wound core in which noise is suppressed.

Means for Solving the Problem

[0008] In order to solve the above problem, the present invention proposes the means described below.

[0009] <1> A wound core according to Aspect 1 of the present invention is: [0010] a wound core formed by laminating, in a sheet thickness direction, a plurality of bent bodies formed from a grain-oriented electrical steel sheet, in which [0011] the wound core has a plurality of flat parts and a plurality of corner portions, [0012] the bent body has a plurality of flat regions and a plurality of bent regions adjacent to the flat regions, [0013] a radius of curvature of each of the bent regions is 5.0 mm or less, [0014] the bent body has one or more joint portions in which end surfaces of the grain-oriented electrical steel sheets in a longitudinal direction face each other, and [0015] when the bent body disposed on the innermost side is defined as a first bent body and a flat region where the joint portion of the first bent body is present is defined as a reference flat region, the joint portion of each of the plurality of bent bodies is located in the flat part having the reference flat region, and [0016] in a side view of the wound core, [0017] when one bent region adjacent to the reference flat region is defined as a first bent region, [0018] the other bent region adjacent to the reference flat region is defined as a second bent region, [0019] an imaginary line passing through an end point of the first bent region on the reference flat region side and parallel to the sheet thickness direction of the reference flat region is defined as a first imaginary line, [0020] an imaginary line passing through an end point of the second bent region on the reference flat region side and parallel to the sheet thickness direction of the reference flat region is defined as a second imaginary line, [0021] among the joint portions of the flat part having the reference flat region, the joint portion located between the first imaginary line and the second imaginary line and having the shortest length from the first imaginary line to the end surface of the joint portion on the first imaginary line side along the longitudinal direction of the reference flat region is defined as a first shortest joint portion, [0022] among the joint portions in the bent bodies adjacent in the sheet thickness direction to the bent body having the first shortest joint portion, the joint portion located between the first imaginary line and the second imaginary line and having a shorter length from the first imaginary line to the end surface of the joint portion on the first imaginary line side along the longitudinal direction of the reference flat region is defined as a first end joint portion, [0023] among the joint portions of the flat part having the reference flat region, the joint portion located between the first imaginary line and the second imaginary line and having the shortest length from the second imaginary line to the end surface of the joint portion on the second imaginary line side along the longitudinal direction of the reference flat region is defined as a second shortest joint portion, [0024] among the joint portions in the bent body adjacent in the sheet thickness direction to the bent body having the second shortest joint portion, the joint portion located between the first imaginary line and the second imaginary line and having a shorter length from the second imaginary line to the end surface of the joint portion on the second imaginary line side along the longitudinal direction of the reference flat region is defined as a second end joint portion, [0025] an imaginary line passing through the end surface of the first shortest joint portion on the first imaginary line side and parallel to the sheet thickness direction of the reference flat region is defined as an imaginary line A, [0026] an imaginary line passing through the end surface of the first end joint portion on the first imaginary line side and parallel to the sheet thickness direction of the reference flat region is defined as an imaginary line B, [0027] an imaginary line passing through the end surface of the second shortest joint portion on the second imaginary line side and parallel to the sheet thickness direction of the reference flat region is defined as an imaginary line C, [0028] an imaginary line passing through the end surface of the second end joint portion on the second imaginary line side and parallel to the sheet thickness direction of the reference flat region is defined as an imaginary line D, [0029] among the joint portions of the flat part having the reference flat region, the joint portion located between the imaginary line A and the imaginary line B is defined as a first-group joint portion, [0030] among the joint portions of the flat part having the reference flat region, the joint portion located between the imaginary line C and the imaginary line D is defined as a second-group joint portion, [0031] an average of lengths from the first imaginary line to the end surface of each of the first-group joint portions on the first imaginary line side along the longitudinal direction of the reference flat region is defined as <L.sub.i>, and [0032] an average of lengths from the second imaginary line to the end surface of each of the second-group joint portions on the second imaginary line side along the longitudinal direction of the reference flat region is defined as <L.sub.O>, [0033] the wound core satisfies the following expressions (1) and (2).

[00001] $2\text{mm} \leq$ .Math. L_i .Math. $< 25\text{mm}$ (1)

$1.22 *$ .Math. Li .Math. $\leq$ .Math. L_O .Math. (2)

[0034] <2> According to Aspect 2 of the present invention, in the wound core of Aspect 1, [0035] the number of first-group joint portions may be equal to the number of second-group joint portions, and [0036] among a quotient and a remainder obtained by dividing the number of joint portions in the flat part located between the first imaginary line and the second imaginary line and having the reference flat region by the number of first-group joint portions, k, which is the quotient, may satisfy the following expression (3).

[00002] $9 \leq k \leq 20$ (3)

[0037] <3> According to Aspect 3 of the present invention, in the wound core of Aspect 1 or 2, [0038] the bent bodies may have the joint portion in each of two flat regions facing each other, [0039] the first bent body may have the reference flat region and a second reference flat region facing the reference flat region, and [0040] the joint portion of each of the plurality of bent bodies may be located in the flat part having the reference flat region and the flat part having the second reference flat region, and [0041] in a side view of the wound core, [0042] when one bent region adjacent to the second reference flat region is defined as a third bent region, [0043] the other bent region adjacent to the second reference flat region is defined as a fourth bent region, [0044] an imaginary line passing through an end point of the third bent region on the second reference flat region side and parallel to the sheet thickness direction of the second reference flat region is defined as a third imaginary line, [0045] an imaginary line passing through an end point of the fourth bent region on the second reference flat region side and parallel to the sheet thickness direction of the second reference flat region is defined as a fourth imaginary line, [0046] among the joint portions of the flat part having the second reference flat region, the joint portion located between the third imaginary line and the fourth imaginary line and having the shortest length from the third imaginary line to the end surface of the joint portion on the third imaginary line side along the longitudinal direction of the second reference flat region is defined as a third shortest joint portion, [0047] among the joint portions in the bent bodies adjacent in the sheet thickness direction to the bent body having the third shortest joint portion, the joint portion located between the third imaginary line and the fourth imaginary line and having a shorter length from the third imaginary line to the end surface of the joint portion on the third imaginary line side along the longitudinal direction of the second reference flat region is defined as a third end joint portion, [0048] among the joint portions of the flat part having the second reference flat region, the joint portion located between the third imaginary line and the fourth imaginary line and having the shortest length from the fourth imaginary line to the end surface of the joint portion on the fourth imaginary line side along the longitudinal direction of the second reference flat region is defined as a fourth shortest joint portion, [0049] among the joint portions in the bent body adjacent in the sheet thickness direction to the bent body having the fourth shortest joint portion, the joint portion located between the third imaginary line and the fourth imaginary line and having a shorter length from the fourth imaginary line to the end surface of the joint portion on the fourth imaginary line side along the longitudinal direction of the second reference flat region is defined as a fourth end joint portion, [0050] an imaginary line passing through the end surface of the third shortest joint portion on the third imaginary line side and parallel to the sheet thickness direction of the second reference flat region is defined as an imaginary line E, [0051] an imaginary line passing through the end surface of the third end joint portion on the third imaginary line side and parallel to the sheet thickness direction of the second reference flat region is defined as an imaginary line F, [0052] an imaginary line passing through the end surface of the fourth shortest joint portion on the fourth imaginary line side and parallel to the sheet thickness direction of the second reference flat region is defined as an imaginary line G, [0053] an imaginary line passing through the end surface of the fourth end joint portion on the fourth imaginary line side and parallel to the sheet thickness direction of the second reference flat region is defined as an imaginary line H, [0054] among the joint portions of the flat part having the second reference flat region, the joint portion located between the imaginary line E and the imaginary line F is defined as a third-group joint portion, [0055] among the joint portions of the flat part having the second reference flat region, the joint portion located between the imaginary line G and the imaginary line H is defined as a fourth-group joint portion, [0056] an average of lengths from the third imaginary line to the end surface of each of the third-group joint portions on the third imaginary line side along the longitudinal direction of the second reference flat region is defined as <L.sub.2i>, and [0057] an average of lengths from the fourth imaginary line to the end surface of each of the fourth-group joint portions on the fourth imaginary line side along the longitudinal direction of the second reference flat region is defined as <L.sub.2O>, [0058] the wound core may satisfy the following expressions (4) and (5).

[00003] $2\mathrm{mm} \leq .\mathrm{Math}.\ L_{2i}\ .\mathrm{Math}. < 25\mathrm{mm}$ (4)

$1.22 * .\mathrm{Math}.\ L_{2i}\ .\mathrm{Math}. \leq .\mathrm{Math}.\ L_{2O}\ .\mathrm{Math}.$ (5)

[0059] <4> According to Aspect 4 of the present invention, in the wound core of Aspect 3, [0060] the number of third-group joint portions may be equal to the number of fourth-group joint portions, and [0061] among a second quotient and a second remainder obtained by dividing the number of joint portions in the flat part located between the third imaginary line and the fourth imaginary line and having the second reference flat region by the number of third-group joint portions, k2, which is the second quotient, may satisfy the following expression (6).

[00004] $9 \leq k2 \leq 20 \quad (6)$

[0062] <5> According to Aspect 5 of the present invention, in the wound core according to any one of Aspects 1 to 4, a bending angle of the bent region may be 30° to 60°.

Effects of the Invention

[0063] According to the above aspects of the present disclosure, it is possible to provide a wound core in which noise is suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0064] FIG. **1** is a perspective view illustrating a wound core according to a first aspect.

[0065] FIG. **2** is a side view of the wound core in FIG. **1**.

[0066] FIG. **3** is a side view illustrating a wound core according to a second aspect.

[0067] FIG. **4** is a side view illustrating a wound core according to a third aspect.

[0068] FIG. **5** is a side view illustrating a wound core according to a fourth aspect.

[0069] FIG. **6** is an enlarged side view of the vicinity of a corner portion of the wound core in FIG. **1**.

[0070] FIG. **7** is an enlarged side view of an example of a bent region.

[0071] FIG. **8** is a side view of a bent body of the wound core in FIG. **1**.

[0072] FIG. **9** is a side view of a wound core of a fifth aspect.

[0073] FIG. **10** is a side view of a wound core of a sixth aspect.

[0074] FIG. **11** is a side view of a wound core of a seventh aspect.

[0075] FIG. **12** is an explanatory view illustrating a first example of a wound core manufacturing apparatus used in a wound core manufacturing method.

[0076] FIG. **13** is a schematic view illustrating dimensions of a wound core manufactured at the time of characteristic evaluation.

EMBODIMENTS OF THE INVENTION

(Wound Core)

[0077] Hereinafter, the wound core of the present disclosure will be described. Note that a numerical range described below includes the lower limit and the upper limit. A numerical value indicated as “more than” or “less than” is not included in the numerical range. In addition, unless otherwise specified, the unit “%” regarding the chemical composition means “mass %”.

[0078] Terms such as “parallel”, “perpendicular”, “identical”, and “at right angle”, values of length and angle, and the like, which specify shapes, geometric conditions, and degrees thereof, used in the present specification are not to be bound by a strict meaning but are to be interpreted including a range in which similar functions can be expected. In the present disclosure, substantially 90° allows an error of +3°, and means a range of 87° to 93°.

[0079] The wound core according to the present disclosure is a wound core formed by laminating, in a sheet thickness direction, a plurality of bent bodies formed from a grain-oriented electrical steel sheet. The grain-oriented electrical steel sheet used for the wound core is preferably a coated grain-oriented electrical steel sheet, in which a coating is formed on at least one surface of the grain-oriented electrical steel sheet. Also, in the case of a coated grain-oriented electrical steel sheet, the wound core according to the present disclosure is preferably a wound core formed by

laminating, in a sheet thickness direction, a plurality of bent bodies formed from a grain-oriented electrical steel sheet such that the coating of the grain-oriented electrical steel sheet is on an outer side.
[0080] The bent body of the wound core of the present disclosure has a flat region and a bent region adjacent to the flat region. Moreover, the bent body of the wound core of the present disclosure has one or more joint portions in which end surfaces of the grain-oriented electrical steel sheets in a longitudinal direction face each other. In the following description, a case where the grain-oriented electrical steel sheet is a coated grain-oriented electrical steel sheet will be described, but the present invention is not limited to the following configuration. Hereinafter, each configuration of the wound core of the present disclosure will be described in detail.
"Coated Grain-Oriented Electrical Steel Sheet"
[0081] The coated grain-oriented electrical steel sheet in the present disclosure includes at least a grain-oriented electrical steel sheet (sometimes referred to as a "base steel sheet" in the present disclosure) and a coating formed on at least one surface of the base steel sheet.
[0082] The coated grain-oriented electrical steel sheet has at least a primary coating as the coating, and may further have another layer as necessary. Examples of the other layer include a secondary coating provided on the primary coating.
[0083] Hereinafter, the configuration of the coated grain-oriented electrical steel sheet will be described.
<Grain-Oriented Electrical Steel Sheet>
[0084] In the coated grain-oriented electrical steel sheet constituting the wound core **10** according to the present disclosure, the base steel sheet is a steel sheet in which the orientation of grains is highly accumulated in a {110}<001> orientation. The base steel sheet has excellent magnetic properties in a rolling direction.
[0085] The base steel sheet used for the wound core according to the present disclosure is not particularly limited. As the base steel sheet, a known grain-oriented electrical steel sheet can be appropriately selected and used. As the grain-oriented electrical steel sheet, an oriented electrical steel strip described in JIS C **2553**:**2019** can be adopted. Hereinafter, an example of the base steel sheet will be described, but the base steel sheet is not limited to the following example.
[0086] The chemical composition of the base steel sheet is not particularly limited, but for example, it is preferable that the base steel sheet contains, in mass %, Si: 0.8% to 7%, C: more than 0% and 0.085% or less, acid-soluble Al: 0% to 0.065%, N: 0% to 0.012%, Mn: 0% to 1%, Cr: 0% to 0.3%, Cu: 0% to 0.4%, P: 0% to 0.5%, Sn: 0% to 0.3%, Sb: 0% to 0.3%, Ni: 0% to 1%, S: 0% to 0.015%, and Se: 0% to 0.015%, and the remainder is Fe and impurity elements.
[0087] The above chemical composition of the base steel sheet is a preferred chemical component for controlling the crystal orientation to a Goss texture accumulated in the {110}<001> orientation.
[0088] Among the elements other than Fe in the base steel sheet, Si and C are basic elements (essential elements). When the Si content of the base steel sheet is 2.0% or more in mass %, eddy-current loss of the wound core is suppressed, which is preferable. The Si content of the base steel sheet is more preferably 3.0% or more. In addition, when the Si content of the base steel sheet is 5.0% or less in mass %, fracture of the steel sheet is less likely to occur in a hot rolling step and cold rolling, which is preferable. The Si content of the base steel sheet is more preferably 4.5% or less.
[0089] The base steel sheet may contain, as optional elements, acid-soluble Al, N, Mn, Cr, Cu, P, Sn, Sb, Ni, S, and Se. Since these optional elements may be contained depending on the object, the lower limit is 0%. In addition, even if these optional elements are contained as impurity elements, the effects of the present disclosure are not impaired.
[0090] The grain-oriented electrical steel sheet generally undergoes purification annealing during secondary recrystallization. In the purification annealing, an inhibitor-forming element is discharged to the outside of the system. Particularly, for N and S, the concentration remarkably

decreases to 50 ppm or less. Under normal purification annealing conditions, the concentration reaches 9 ppm or less, further 6 ppm or less, and a degree that cannot be detected by general analysis (1 ppm or less) if purification annealing is sufficiently performed.

[0091] In the base steel sheet, the remainder of the basic elements and the optional elements is Fe and impurity elements. Here, the “impurity element” means an element unintentionally mixed from ore as a raw material, scrap, a manufacturing environment, or the like when the base steel sheet is industrially manufactured.

[0092] The chemical component of the base steel sheet may be measured by a general analysis method of steel. For example, the chemical component of the base steel sheet may be measured by inductively coupled plasma-atomic emission spectrometry (ICP-AES). Specifically, for example, the chemical component can be specified by acquiring a test piece of 35 mm square from a center position in a width direction of the base steel sheet after removal of a coating, and performing measurement under a condition based on a calibration curve created in advance using an apparatus (measurement apparatus) such as ICPS-8100 manufactured by Shimadzu Corporation. C and S may be measured by a combustion-infrared absorption method, and N may be measured by an inert gas fusion-thermal conductivity method.

[0093] The chemical component of the base steel sheet is a component obtained by analyzing a component of a steel sheet obtained by removing a glass coating, a coating containing phosphorus, and the like described later from a grain-oriented electrical steel sheet by a method described later as the base steel sheet.

<Primary Coating>

[0094] The primary coating is a coating directly formed on a surface of a grain-oriented electrical steel sheet as a base steel sheet without any other layer or film. Examples of the primary coating include a glass coating. Examples of the glass coating include a coating having one or more oxides selected from forsterite (Mg.sub.2SiO.sub.4), spinel (MgAl.sub.2O.sub.4), and cordierite (Mg.sub.2Al.sub.4Si.sub.5O.sub.16). For example, a coating containing phosphorus described later may be formed as a primary coating without forming a glass coating on a surface of a grain-oriented electrical steel sheet.

[0095] When the primary coating is a glass coating, the method for forming the glass coating is not particularly limited, and can be appropriately selected from known methods. For example, the method includes a method in which an annealing separator containing one or more selected from magnesia (MgO) and alumina (Al.sub.2O.sub.3) is applied to a cold-rolled steel sheet, and then finish annealing is performed.

[0096] The annealing separator also has an effect of suppressing sticking of steel sheets during finish annealing. For example, when finish annealing is performed by applying the annealing separator containing magnesia, silica contained in the base steel sheet reacts with the annealing separator to form a glass coating containing forsterite (Mg.sub.2SiO.sub.4) on a base steel sheet surface.

[0097] The thickness of the primary coating is not particularly limited, but is preferably, for example, 0.5 μm or more and 3 μm or less from the viewpoint of forming the primary coating on the entire surface of a base steel sheet and suppressing peeling.

<Other Coatings>

[0098] The coated grain-oriented electrical steel sheet may include a coating other than the primary coating. For example, it is preferable that the coated grain-oriented electrical steel sheet have a coating containing phosphorus as other film (a secondary coating) on the primary coating. By having a coating containing phosphorus, insulation properties can be improved. The coating containing phosphorus is a coating formed on the outermost surface of the grain-oriented electrical steel sheet. When the grain-oriented electrical steel sheet has a glass coating or an oxide film as a primary coating, the grain-oriented electrical steel sheet is formed on the primary coating. By forming a coating containing phosphorus on the glass coating formed as a primary coating on the

surface of the base steel sheet, high adhesion can be secured.
[0099] The coating containing phosphorus can be appropriately selected from conventionally known coatings. The coating containing phosphorus is preferably a phosphate-based coating, and particularly preferably a coating containing one or more of aluminum phosphate and magnesium phosphate as main components, and further containing one or more of chromium and silicon oxide as accessory components. According to the phosphate-based coating, insulation properties of the steel sheet are secured, and tension is imparted to the steel sheet to be excellent in reduction of iron loss.
[0100] When the other film is a coating containing phosphorus, the thickness of the coating containing phosphorus is not particularly limited, but is preferably 0.5 μm or more and 3 μm or less from the viewpoint of securing insulation properties.
<Sheet Thickness>
[0101] The sheet thickness of the coated grain-oriented electrical steel sheet is not particularly limited, and may be appropriately selected according to the application and the like, but is usually in the range of 0.10 mm to 0.50 mm, preferably 0.13 mm to 0.35 mm, and more preferably in the range of 0.15 mm to 0.30 mm.
(Configuration of Wound Core)
[0102] A configuration of the wound core according to the present disclosure will be described with reference to a wound core **10** in FIGS. **1** and **2** as an example. FIG. **1** is a perspective view of a wound core **10**, and FIG. **2** is a side view of the wound core **10** in FIG. **1**.
[0103] In the present disclosure, viewing from the side means viewing in a width direction (Y-axis direction in FIG. **1**) of a grain-oriented electrical steel sheet in a long shape constituting a wound core.
[0104] The side view is a view illustrating a shape visually recognized by viewing from the side (a view in the Y-axis direction in FIG. **1**). The sheet thickness direction is a sheet thickness direction of the grain-oriented electrical steel sheet. In the wound core **10** of the present disclosure, the sheet thickness direction is a direction perpendicular to the circumferential surface of the wound core in a state of being formed into a rectangular wound core.
[0105] The direction perpendicular to a circumferential surface means a direction perpendicular to the circumferential surface when the circumferential surface is viewed from the side. When the circumferential surface forms a curve in a side view, the direction perpendicular to the circumferential surface (sheet thickness direction) means a direction perpendicular to a tangent of the curve formed by the circumferential surface.
[0106] The wound core **10** is configured by laminating a plurality of bent bodies **1** in a sheet thickness direction thereof. For example, as illustrated in FIGS. **1** and **2**, the wound core **10** has a substantially rectangular laminated structure including a plurality of bent bodies **1**. The wound core **10** has a laminated body **2** obtained by laminating the plurality of bent bodies **1**. The wound core **10** may be used as it is as a wound core. If necessary, the wound core **10** may be fixed using a fastening tool such as a known binding band. The bent body **1** is formed of a grain-oriented electrical steel sheet which is a base steel sheet. The number of bent bodies **1** (the number of laminated sheets) is not particularly limited, but for example, the number of bent bodies **1** is preferably 200 or more.
[0107] As illustrated in FIGS. **1** and **2**, the wound core **10** is preferably formed in a rectangular shape by alternately continuing four flat parts **4** and four corner portions **3** along a circumferential direction. The wound core **10** has a plurality of flat parts **4** and a plurality of corner portions **3**. An angle formed by two flat parts **4** adjacent to each corner portion **3** is preferably substantially 90°. Here, the circumferential direction means a direction around an axis of the wound core **10**.
[0108] At the corner portion **3** of the wound core **10**, the bent body **1** has two bent regions **5** (FIG. **2**). The bent region **5** is a region having a curved bent shape in viewing the bent body **1** from the side. The bent region will be described in detail later. In the two bent regions **5**, bending angles in

total are preferably substantially 90° in viewing the bent body **1** from the side.
[0109] In each of the corner portions **3** of the wound core **10**, the bent body **1** may have one or more bent regions **5** so that the grain-oriented electrical steel sheet is bent by substantially 90°. As in a wound core **10**A according to a second aspect of the present disclosure, in each of the corner portions **3** of the wound core **10**, the bent body **1** may have three bent regions **5** (FIG. **3**). Also, in each of the corner portions **3** of the wound core **10**, the bent body **1** may have one bent region **5** in one corner portion **3** of the wound core **10**, as in a wound core **10**B according to a third aspect (FIG. **4**). Moreover, in each of the corner portions **3** of the wound core **10**, the bent body **1** may have one bent region **5** in one corner portion **3** of the wound core **10**, as in a wound core **10**G according to a fourth aspect (FIG. **5**). Further, as in the wound core **10**G, the lengths of the flat parts **4** facing each other may be different.
(Flat Region)
[0110] As illustrated in FIG. **2**, the bent body **1** has a flat region **8** adjacent to a bent region **5**. As the flat region **8** adjacent to a bent region **5**, there are two flat regions **8** shown in (1A) and (1B) below.
[0111] (1A) A flat region **8** positioned between a bent region **5** and a bent region **5** (between two bent regions **5** adjacent in the circumferential direction) in one corner portion **3** and adjacent to each bent region **5** (a flat region of a corner portion).
[0112] (1B) A flat region **8** adjacent to each bent region **5** as a flat part **4**.
(Corner Portion)
[0113] FIG. **6** is an enlarged side view of the vicinity of a corner portion **3** in the wound core **10** in FIG. **1**.
[0114] As illustrated in FIG. **6**, in one corner portion **3**, when a bent body **1***a* has two bent region **5***a* and bent region **5***b*, the bent region **5***a* (curved portion) is continuous from a flat region **8***a* belonging to the flat part **4** which is a flat region of the bent body **1***a*, and further, a flat region **7***a* (straight portion), the bent region **5***b* (curved portion), and a flat region **8***b* (straight portion) belonging to the flat part **4***b* are continuous therebeyond.
[0115] In the wound core **10**, a region from a line segment A-A′ to a line segment B-B′ in FIG. **6** is the corner portion **3**. A point A is an end point on the flat region **8***a* side in the bent region **5***a* of the bent body (first bent body) **1***a* disposed on the innermost side of the wound core **10**. A point A′ is an intersection point of a straight line passing through the point A and perpendicular (sheet thickness direction) to a sheet surface of the bent body **1***a* and the outermost surface of the wound core **10** (an outer circumferential surface of the bent body **1** disposed on the outermost side of the wound core **10**). Similarly, a point B is an end point on the flat region **8***b* side in the bent region **5***b* of the bent body **1***a* disposed on the innermost side of the wound core **10**. A point B′ is an intersection point of a straight line passing through the point B and perpendicular (sheet thickness direction) to a sheet surface of the bent body **1***a* and the outermost surface of the wound core **10**. In FIG. **6**, an angle formed by two flat parts **4***a* and **4***b* adjacent to each other with the corner portion **3** interposed therebetween (angle formed by intersection of extension lines of the flat parts **4***a* and **4***b*) is θ, and in the example in FIG. **6**, the θ is substantially 90°. The bending angles of the bent regions **5***a* and **5***b* will be described later, but in FIG. **6**, the bending angles in total φ**1**+φ**2** of the bent regions **5***a* and **5***b* are substantially 90°. The bending angle φ**1** of the bent region **5***a* is, for example, 30° to 60°. Similarly, the bending angle φ**2** of the bent region **5***b* is, for example, 30 to 60°. Since the bending angles φ**1** and φ**2** of the bent regions **5***a* and **5***b* are smaller than 90° in the deformation amount, the elastic stress due to bending, that is, bending return becomes small and the variation in angle becomes small, and thus the bending angles φ**1** and φ**2** of the bent regions **5***a* and **5***b* are particularly preferably 30 to 60°.
(Bent Region)
[0116] The bent region **5** will be described in detail with reference to FIG. **7**. FIG. **7** is an enlarged side view of an example of the bent region **5** of the bent body **1**. The bending angle q of the bent

region **5** means an angular difference generated between a flat region on a rear side in a bending direction and a flat region on a front side in the bending direction in the bent region **5** of the bent body **1**. Specifically, the bending angle q of the bent region **5** is represented as an angle q of a complementary angle of an angle formed by two imaginary lines Lb-elongation **1** and Lb-elongation **2** obtained by extending straight portions adjacent to respective points from points (points F and G) on both sides of a curved portion included in a line Lb representing an outer surface of the bent body **1** in the bent region **5**.

[0117] The bending angle of each bent region **5** is preferably substantially 90° or less, and the bending angles in total of all the bent regions **5** of the bent body **1** existing in one corner portion **3** of the wound core **10** are substantially 90°.

[0118] In viewing the bent body **1** from the side, when points D and E on a line La representing an inner surface of the bent body **1** and the points F and G on the line Lb representing the outer surface of the bent body **1** are defined as follows, the bent region **5** indicates a region surrounded by (2A) a line delimited by the point D and the point E on the line La representing the inner surface of the bent body **1**, (2B) a line delimited by the point F and the point G on the line Lb representing the outer surface of the bent body **1**, (2C) a straight line connecting the point D and the point G, and (2D) a straight line connecting the point E and the point F.

[0119] Here, the point D, the point E, the point F, and the point G are defined as follows.

[0120] In viewing from the side, a point at which a straight line AB connecting a center point A of a radius of curvature in a curved portion included in the line La representing the inner surface of the bent body **1** and an intersection point B of the two imaginary lines Lb-elongation **1** and Lb-elongation **2** obtained by extending straight portions adjacent to both sides of the curved portion included in the line Lb representing the outer surface of the bent body **1** intersects the line La representing the inner surface of the bent body **1** is defined as an origin C, [0121] a point separated from the origin C, for example, by a distance m represented by the following formula (A) in one direction along the line La representing the inner surface of the bent body **1** is defined as the point D, [0122] a point separated from the origin C, for example, by the distance m in another direction along the line La representing the inner surface of the bent body is defined as the point E, [0123] an intersection point between a straight portion facing the point D among the straight portions included in the line Lb representing the outer surface of the bent body and an imaginary line drawn perpendicularly to the straight portion facing the point D and passing through the point D is defined as the point G, and [0124] an intersection point between a straight portion facing the point E among the straight portions included in the line Lb representing the outer surface of the bent body and an imaginary line drawn perpendicularly to the straight portion facing the point E and passing through the point E is defined as the point F. The intersection point A is an intersection point obtained by extending a line segment EF and a line segment DG inward on the opposite side of the point B.

[00005] $m = r \times (\quad \times \quad / 180) \quad (A)$

[0125] In formula (A), m represents a distance from the origin C, and r represents a distance (radius of curvature) from the center point A to the origin C. The radius of curvature r of the bent body **1** disposed on an inner surface side of the wound core **10** is preferably, for example, 1 mm or more and 5 mm or less. Here, the radius of curvature of the bent body **1** is the radius of curvature of the bent region **5**. The radius of curvature of the bent body **1** is 5.0 mm or less. When the radius of curvature of the bent body **1** is 5.0 mm or less, noise is improved. The radius of curvature of the bent body **1** is preferably 0.1 mm or more. The radius of curvature of the bent body **1** is further preferably 0.3 mm or more. A particularly preferred radius of curvature of the bent body is 1.0 mm or more. A more preferred radius of curvature of the bent body **1** is 2.9 mm or less.

[0126] FIG. **8** is a side view of the bent body **1** of the wound core **10** in FIG. **1**. As illustrated in FIG. **8**, the bent body **1** is obtained by bending a grain-oriented electrical steel sheet, and has a flat region **8** and a bent region **5** adjacent to the flat region **8**. The bent body **1** has a plurality of flat regions **8** and a plurality of bent regions **5**. Also, the bent body **1** has four bent body corner portions **30** and four bent body flat parts **40**, so that one grain-oriented electrical steel sheet forms a substantially rectangular ring in viewing from the side. More specifically, one bent body flat part **40** is provided with a gap (joint portion) **6** in which both end surfaces in the longitudinal direction of the grain-oriented electrical steel sheet face each other, and the other three bent body flat parts **40** have one or more joint portions in which the end surfaces **13** and **14** in the longitudinal direction of the bent body **1** face each other in the joint portion **6** of the bent body **1** having a structure not including the gap **6**. The size of the gap of the joint portion **6** is, for example, 0.1 mm to 5.0 mm, and desirably 1.0 mm to 2.0 mm.

[0127] The wound core **10** preferably has a laminated structure having a substantially rectangular shape as a whole in viewing from the side. The wound core **10** may have a configuration in which two bent body flat parts **40** include the gap (joint portion) **6** and the other two bent body flat parts **4** do not include the gap **6**. In this case, a bent body is formed of two grain-oriented electrical steel sheets.

[0128] It is desirable to prevent generation of a gap between two adjacent layers in a sheet thickness direction at the time of manufacturing the wound core. Therefore, in the two adjacent bent bodies, the length of the steel sheet and the position of the bent region are adjusted such that an outer circumferential length of a bent body flat part **40** of a bent body disposed inside is equal to an inner circumferential length of a bent body flat part **40** of a bent body disposed outside.

(Arrangement of Joint Portions)

[0129] As illustrated in FIG. **2**, when the bent body disposed on the innermost side is defined as a first bent body **1***a* and a flat region where the joint portion **6** of the first bent body **1***a* is present is defined as a reference flat region **11**, a joint portion **6** of each of the plurality of bent bodies **1** is in a flat part **4** having the reference flat region **11**. With such a configuration, windings can be easily assembled.

(Distance of First-Group Joint Portion and Distance of Second-Group Joint Portion)

[0130] In the wound core **10**, a joint portion **6** is arranged such that an average distance of first-group joint portions <L.sub.i> described later and an average distance of second-group joint portions <L.sub.O> described later, which are present near the corner portions **3**, satisfy the following expressions (1) and (2).

[0131] Plastic strain and elastic strain are introduced in the bent region **5**, and strain due to shearing is introduced in an end portion of the joint portion **6**. Noise is generated by extension and contraction of the grain-oriented electrical steel sheet during AC excitation. In particular, in the case of a grain-oriented electrical steel sheet into which strain is introduced, noise is significantly deteriorated. In the wound core **10**, when the average distance of first-group joint portions <L.sub.i> and the average distance of second-group joint portions <L.sub.O> described later satisfy the following expressions (1) and (2), the strain affected region in the bent region **5** and the shear strain affected region in the vicinity of the joint portion **6** can be densely arranged. This makes it possible to reduce the strain affected region in the entire wound core **10**. As a result, noise can be reduced. The plastically deformed grain-oriented electrical steel sheet is cured by strain. Therefore, if the joint portion **6** is formed by shearing the grain-oriented electrical steel sheet near the bent region **5**, burrs generated at the time of shearing may lead to coating damage of other laminated grain-oriented electrical steel sheets. In addition, even if folding is performed after shearing, shape defects of bending angle and the like occur. Furthermore, plastic strain and elastic strain interfere with strain due to shearing in the end portion of the joint portion **6** in the bent region **5**, so that iron loss is further deteriorated. Therefore, <L.sub.i> is preferably 2 mm or more. In the first-group joint portion and the second-group joint portion, a joint portion with a smaller average distance is defined as the first-group joint portion.

[00006] $2\text{mm} \leq$ .Math. L_i .Math. $< 25\text{mm}$ (1)

$1.22 * L_i \leq L_O$ (2)

(First-Group Joint Portion V.SUB.i.)

[0132] Next, a first-group joint portion V.sub.i and a second-group joint portion V.sub.O will be described by taking a case where there are a plurality of first-group joint portions V.sub.i and a plurality of second-group joint portions as an example. FIG. **9** is a side view of a wound core **10**D of a fifth aspect having a plurality of first-group joint portions V.sub.i and a plurality of second-group joint portions V.sub.O. A plurality of bent bodies **1** are also laminated on a portion " . . . " between a bent body **1** and a bent body **1** of the wound core **10**D in FIG. **9**. The wound core **10**D is a wound core in which the bent body **1** having one joint portion **6** is laminated. In FIG. **9**, the bent body disposed on the innermost side is defined as a first bent body **1***a*, and a flat region where a joint portion **6** of the first bent body **1***a* is present is defined as a reference flat region **11**. In the wound core **10**D, each joint portion **6** is located in a flat part **4** having a reference flat region **11**. In FIG. **9**, the flat part **4** where the joint portion **6** is present is a flat part parallel to the X direction.

[0133] Also, one bent region adjacent to the reference flat region **11** is defined as a first bent region **12***a*, and the other bent region adjacent to the reference flat region **11** is defined as a second bent region **12***b*. An imaginary line passing through an end point of the first bent region **12***a* on the reference flat region **11** side and parallel to the sheet thickness direction of the reference flat region **11** is defined as a first imaginary line H**1**, and an imaginary line passing through an end point of the second bent region **12***b* on the reference flat region **11** side and parallel to the sheet thickness direction of the reference flat region **11** is defined as a second imaginary line H**2**.

[0134] Among the joint portions **6** of the flat part **4** having the reference flat region **11**, a joint portion **6** located between the first imaginary line H**1** and the second imaginary line H**2** and having the shortest length from the first imaginary line H**1** to the end surface **13** of the joint portion **6** on the first imaginary line H**1** side along the longitudinal direction of the reference flat region **11** is defined as a first shortest joint portion **6***a*. Among the joint portions **6** in bent bodies **1***c* and **1***d* adjacent in the sheet thickness direction to the bent body **1***b* having the first shortest joint portion **6***a*, a joint portion **6** located between the first imaginary line H**1** and the second imaginary line H**2** and having a shorter length from the first imaginary line H**1** to the end surface **13** of the joint portion **6** on the first imaginary line H**1** side along the longitudinal direction of the reference flat region **11** is defined as a first end joint portion **6***b*.

[0135] An imaginary line passing through an end surface **13***a* of the first shortest joint portion **6***a* on the first imaginary line H**1** side and parallel to the sheet thickness direction of the reference flat region **11** is defined as an imaginary line A. An imaginary line passing through an end surface **13***b* of the first end joint portion **6***b* on the first imaginary line H**1** side and parallel to the sheet thickness direction of the reference flat region **11** is defined as an imaginary line B. Among the joint portions **6** of the flat part **4** having the reference flat region **11**, a joint portion **6** located between the imaginary line A and the imaginary line B is defined as the first-group joint portion V.sub.i. Here, the number of first-group joint portions V.sub.i is n (n is a natural number) in total from V.sub.i to V.sub.in.

(Average Distance of First-Group Joint Portions <L.SUB.i.>)

[0136] An average of lengths from the first imaginary line H**1** to the end surface of each of the first-group joint portions V.sub.i on the first imaginary line H**1** side along the longitudinal direction of the reference flat region **11** is defined as an average distance of first-group joint portions V.sub.i<L.sub.i>. The average distance of first-group joint portions V.sub.i <L.sub.i> can be measured by the following method. An observation image of the side surface of the wound core is obtained using an optical microscope or the like. In the obtained observation image, the first-group joint portion is specified based on the definition described above. Next, length L.sub.i from the first imaginary line H**1** to the end surface of the first-group joint portion V.sub.i on the first imaginary line H**1** side along the longitudinal direction of the reference flat region **11** is measured using image processing software. An average value of the obtained L.sub.i is obtained, and the average value is defined as the average distance of first-group joint portions <L.sub.i>.

(Second-Group Joint Portion V.SUB.O.)

[0137] Next, the second-group joint portion V.sub.O will be described. Among the joint portions **6** of the flat part **4** having the reference flat region **11**, a joint portion located between the first imaginary line H**1** and the second imaginary line H**2** and having the shortest length from the second imaginary line H**2** to the end surface **14** of the joint portion **6** on the second imaginary line H**2** side along the longitudinal direction of the reference flat region **11** is defined as a second shortest joint portion **6***c*. Among the joint portions **6** in bent bodies **1***f* and **1***g* adjacent in the sheet thickness direction to the bent body **1***e* having the second shortest joint portion **6***c*, a joint portion **6** located between the first imaginary line H**1** and the second imaginary line H**2** and having a shorter length from the second imaginary line H**2** to the end surface **14** of the joint portion **6** on the second imaginary line H**2** side along the longitudinal direction of the reference flat region **11** is defined as a second end joint portion **6***d*.

[0138] An imaginary line passing through an end surface **14***a* of the second shortest joint portion **6***c* on the second imaginary line H**2** side and parallel to the sheet thickness direction of the reference flat region **11** is defined as an imaginary line C. An imaginary line passing through an end surface **14***b* of the second end joint portion **6***d* on the second imaginary line H**2** side and parallel to the sheet thickness direction of the reference flat region **11** is defined as an imaginary line D. Among the joint portions **6** of the flat part **4** having the reference flat region **11**, a joint portion **6** located between the imaginary line C and the imaginary line C is defined as a second-group joint portion V.sub.O. Here, the number of second-group joint portions V.sub.O is m (m is a natural number) in total from V.sub.O1 to V.sub.Om.

(Average Distance of Second-Group Joint Portions <L.SUB.O.>)

[0139] An average of lengths from the second imaginary line H**2** to the end surface of each of the second-group joint portions V.sub.O on the second imaginary line H**2** side along the longitudinal direction of the reference flat region **11** is defined as an average distance of second-group joint portions V.sub.O<L.sub.O>. The average distance of second-group joint portions V.sub.O<L.sub.O> can be measured by the following method. An observation image of the side surface of the wound core is obtained using an optical microscope or the like. In the obtained observation image, the second-group joint portion V.sub.O is specified based on the definition described above. Next, length L.sub.O from the second imaginary line H**2** to the end surface of the second-group joint portion V.sub.O on the second imaginary line H**2** side along the longitudinal direction of the reference flat region **11** is measured using image processing software. An average value of the obtained L.sub.O is obtained, and the average value is defined as the average distance of second-group joint portions <L.sub.O>.

[0140] In the wound core **10**D, the joint portions **6** are preferably arranged such that the joint portions **6** are shifted from each other in a stepwise manner in the circumferential direction. In the wound core **10**D, the circumferential direction is the same as the longitudinal direction of the reference flat region **11**. The circumferential position of the joint portion **6** in the bent body **1** is gradually shifted from the first imaginary line H**1** side (first-group joint portion Vi side) to the second imaginary line H**2** side (second-group joint portion Vo side) in the circumferential direction from the bent body **1** located on the inner side in the radial direction toward the bent body **1** located on the outer side in the radial direction. The radial direction refers to a direction orthogonal to the axis of the wound core **10**D. Hereinafter, such a pattern of arrangement of the joint portions **6** is referred to as a stepwise pattern. In the present embodiment, the joint portions **6** are arranged such that a plurality of stepwise patterns are repeated in the radial direction. In a first embodiment, among the joint portions **6** arranged in one stepwise pattern, a joint portion **6** of the bent body **1** located on the innermost side in the radial direction is included in the first-group joint portion Vi, and a joint portion **6** of the bent body **1** located on the outermost side in the radial direction is included in the second-group joint portion Vo. By sequentially shifting the joint portions **6** along the circumferential direction in this manner, it is possible to suppress inhibition of a flow of magnetic flux in the wound core **10**D.

[0141] In the wound core **10**D, the number of first-group joint portions V.sub.i is preferably equal to the number of second-group joint portions V.sub.O. Also, in the wound core **10**D, among a quotient and a remainder obtained by dividing the number of joint portions **6** in the flat part **4** located between the first imaginary line H**1** and the second imaginary line H**2** and having the reference flat region **11** by the number of first-group joint portions V.sub.i, the quotient is defined as k, and k satisfies the following expression (3). In FIG. **9**, this number k is equal to the number of joint portions located between V.sub.i1 and V.sub.o1 and located between the first imaginary line H**1** and the second imaginary line H**2** along the sheet thickness direction. That is, k is the number of joint portions **6** arranged to be shifted stepwise from the first-group joint portion V.sub.i to the second-group joint portion V.sub.O closest to the first-group joint portion V.sub.i. The number k is the number of joint portions included in one stepwise pattern. By arranging the joint portion **6** in this manner, noise can be further suppressed.

[00007] $9 \leq k \leq 20 \quad (3)$

(Length Between Imaginary Line a and Imaginary Line C)

[0142] A length between the imaginary line A and the imaginary line C along the longitudinal direction is preferably 50% or more of a length between the first imaginary line H**1** and the second imaginary line H**2** along the longitudinal direction. That is, (the length between the imaginary line A and the imaginary line C along the longitudinal direction)/(the length between the first imaginary line H**1** and the second imaginary line H**2** along the longitudinal direction)×100 is 50% or more. Since the length between the imaginary line A and the imaginary line C along the longitudinal direction is 50% or more of the length between the first imaginary line H**1** and the second imaginary line H**2** along the longitudinal direction, noise can be further suppressed. More preferably, the length between the imaginary line A and the imaginary line C along the longitudinal direction is 60% or more of the length between the first imaginary line H**1** and the second imaginary line H**2** along the longitudinal direction.

[0143] In FIG. **9**, the flat part **4** where the joint portion **6** is present is a flat part parallel to the X direction, but the position of the joint portion in the present invention is not limited to the configuration of FIG. **9**. For example, as in a wound core **10**E of the sixth aspect in FIG. **10**, the flat part **4** where the joint portion **6** is present may be a flat part parallel to the Z direction.

[0144] In the wound core **10**E, the average distance of first-group joint portions V.sub.i<L.sub.i> and the average distance of second-group joint portions V.sub.O<L.sub.O> satisfy the above expressions (1) and (2). When the average distance of first-group joint portions <L.sub.i> and the average length <L.sub.O> satisfy the above expressions (1) and (2), noise can be suppressed.

[0145] In the wound core **10**E, the joint portions **6** are preferably arranged such that the joint portions **6** are shifted from each other in a stepwise manner in the circumferential direction. By sequentially shifting the joint portions **6** along the circumferential direction in this manner, it is possible to suppress inhibition of a flow of magnetic flux in the wound core **10**E.

[0146] In the wound core **10**E, similarly to the wound core **10**D, the number of first-group joint portions V.sub.i is preferably equal to the number of second-group joint portions V.sub.O. Also, in the wound core **10**E, the number k obtained by dividing the number of joint portions **6** in the flat part **4** located between the first imaginary line H**1** and the second imaginary line H**2** and having the reference flat region **11** by the number of first-group joint portions V.sub.i satisfies the above expression (3). It is preferable that by arranging the joint portion **6** in this manner, noise can be further suppressed.

[0147] In FIGS. **9** and **10**, the example of the bent body **1** having one joint portion **6** has been described, but the number of joint portions is not limited to one in the present invention. For example, as in a wound core **10**F of the seventh aspect in FIG. **11**, each bent body **1** may have a joint portion **6** in each of two flat regions **8** facing each other When each bent body **1** has two joint portions **6**, a first bent body **1***a* of the wound core **10**F has a reference flat region **11** and a second reference flat region **11***b* facing the reference flat region **11**.

(Distance of Third-Group Joint Portion and Distance of Fourth-Group Joint Portion)

[0148] In the wound core **10**F, a joint portion **6** is preferably arranged such that an average distance of third-group joint portions <L.sub.2i> described later and an average distance of fourth-group joint portions <L.sub.2O> described later, which are present near the corner portions **3**, satisfy the following expressions (4) and (5).

[0149] Plastic strain and elastic strain are introduced in the bent region **5**, and strain due to shearing is introduced in an end portion of the joint portion **6**. Noise is generated by extension and contraction of the grain-oriented electrical steel sheet during AC excitation. In particular, in the case of a grain-oriented electrical steel sheet into which strain is introduced, noise is significantly deteriorated. In the wound core **10**, when the average distance of third-group joint portions <L.sub.2i> and the average distance of fourth-group joint portions <L.sub.2O> described later satisfy the following expressions (4) and (5), the strain affected region in the bent region **5** and the shear strain affected region in the vicinity of the joint portion **6** can be densely arranged. This makes it possible to further reduce the strain affected region in the entire wound core **10**. As a result, noise can be further reduced. The plastically deformed grain-oriented electrical steel sheet is cured by strain. Therefore, if the joint portion **6** is formed by shearing the grain-oriented electrical steel sheet near the bent region **5**, burrs generated at the time of shearing may lead to coating damage of other laminated grain-oriented electrical steel sheets. In addition, even if folding is performed after shearing, shape defects of bending angle and the like occur. Furthermore, plastic strain and elastic strain interfere with strain due to shearing in the end portion of the joint portion **6** in the bent region **5**, so that iron loss is further deteriorated. Therefore, the lower limit of <L.sub.2i> in the following expression (4) is preferably 2 mm. In the third-group joint portion and the fourth-group joint portion, a joint portion with a smaller average distance is defined as the third-group joint portion. When there are two joint portions **6** in the bent body **1** constituting the wound core and only the plurality of joint portions **6** of one flat part **4** of two flat parts **4** where the joint portions **6** are present satisfy the above expressions (1) and (2), a flat part **4** where the plurality of joint portions **6** satisfying the above expressions (1) and (2) are present is defined as a flat part **4***c* having the reference flat region **11**.

[00008] $2\text{mm} \leq\ .\text{Math.}\ L_{2i}\ .\text{Math.}\ < 25\text{mm}$ (4)

$1.22 \times\ .\text{Math.}\ L_{2i}\ .\text{Math.}\ \leq\ .\text{Math.}\ L_{2O}\ .\text{Math.}$ (5)

(Third-Group Joint Portion V.SUB.2i.)

[0150] Next, a third-group joint portion V.sub.2i and a fourth-group joint portion V.sub.2o will be described by taking the wound core **10**F in FIG. **11** as an example. Description of the plurality of first-group joint portions V.sub.i and the plurality of second-group joint portions V.sub.O will be omitted. FIG. **11** is a side view of a wound core **10**F having a plurality of third-group joint portions V.sub.2i and a plurality of fourth-group joint portions V.sub.2O. The wound core **10**F is a wound core in which the bent body **1** having one joint portion **6** is laminated. In FIG. **11**, the bent body disposed on the innermost side is defined as a first bent body **1***a*. The first bent body **1***a* has a reference flat region **11** and a second reference flat region **11***b*. The second reference flat region **11***b* is a flat region facing the reference flat region **11**, and has a joint portion **6**. The joint portion **6** of each of the plurality of bent bodies **1** is located in the flat part **4***c* having the reference flat region **11** and a flat part **4***d* having the second reference flat region **11***b*. In FIG. **11**, the flat parts **4***c* and **4***d* where the joint portion **6** is present are flat parts parallel to the X direction.

[0151] One bent region adjacent to the second reference flat region **11***b* is defined as a third bent region **12***c*, and the other bent region adjacent to the second reference flat region **11***b* is defined as a fourth bent region 12d. An imaginary line passing through an end point of the third bent region 12c on the second reference flat region 11b side and parallel to the sheet thickness direction of the second reference flat region 11b is defined as a third imaginary line H1a, and an imaginary line passing through an end point of the fourth bent region 12d on the second reference flat region 11b side and parallel to the sheet thickness direction of the second reference flat region 11b is defined as a fourth imaginary line H2a.

[0152] Among the joint portions 6 of the flat part 4d having the second reference flat region 11b, a joint portion 6 located between the third imaginary line H1a and the fourth imaginary line H2a and having the shortest length from the third imaginary line H1a to the end surface 13 of the joint portion 6 on the third imaginary line H1a side along the longitudinal direction of the second reference flat region 11b is defined as a third shortest joint portion 6e. Among the joint portions 6 in bent bodies 1i and 1j adjacent in the sheet thickness direction to the bent body 1h having the third shortest joint portion 6e, a joint portion 6 located between the third imaginary line H1a and the fourth imaginary line H2a and having a shorter length from the third imaginary line H1a to the end surface 13 of the joint portion 6 on the third imaginary line H1a side along the longitudinal direction of the second reference flat region 11b is defined as a third end joint portion 6f.

[0153] An imaginary line passing through an end surface 13c of the third shortest joint portion 6e on the third imaginary line H1a side and parallel to the sheet thickness direction of the second reference flat region 11b is defined as an imaginary line E. An imaginary line passing through an end surface 13d of the third end joint portion 6f on the third imaginary line H1a side and parallel to the sheet thickness direction of the second reference flat region 11b is defined as an imaginary line F. Among the joint portions 6 of the flat part 4d having the second reference flat region 11b, a joint portion 6 located between the imaginary line E and the imaginary line F is defined as the third-group joint portion V.sub.2i. Here, the number of third-group joint portions V.sub.2i is n (n is a natural number) in total from V.sub.2i to V.sub.2in.

(Average Distance of Third-Group Joint Portions <L.SUB.2i.>)

[0154] An average of lengths from the third imaginary line H1a to the end surface 13 of the third-group joint portions V.sub.2i on the third imaginary line H1a side along the longitudinal direction of the second reference flat region 11b is defined as an average distance of third-group joint portions V.sub.2i <L.sub.2i>. The average distance of third-group joint portions V.sub.2i <L.sub.2i> can be measured by the following method. An observation image of the side surface of the wound core is obtained using an optical microscope or the like. In the obtained observation image, the third-group joint portion is specified based on the definition described above. Next, length L.sub.2i from the third imaginary line H1a to the end surface 13 of the third-group joint portion V.sub.2i on the third imaginary line H1a side along the longitudinal direction of the second reference flat region 11b (flat region facing the first reference flat region) is measured using image processing software. An average value of the obtained L.sub.i is obtained, and the average value is defined as the average distance of third-group joint portions <L.sub.2i>.

(Fourth-Group Joint Portion V.SUB.2.o)

[0155] Next, the fourth-group joint portion V.sub.2o will be described. Among the joint portions 6 of the flat part 4d having the second reference flat region 11b, a joint portion 6 located between the third imaginary line H1a and the fourth imaginary line H2a and having the shortest length from the fourth imaginary line H2a to the end surface 14 of the joint portion 6 on the fourth imaginary line H2a side along the longitudinal direction of the second reference flat region 11b is defined as a fourth shortest joint portion 6g. Among the joint portions 6 in bent bodies 1l and 1m adjacent in the sheet thickness direction to the bent body 1k having the fourth shortest joint portion 6g, a joint portion 6 located between the third imaginary line H1a and the fourth imaginary line H2a and having a shorter length from the fourth imaginary line H2a to the end surface 14 of the joint portion 6 on the fourth imaginary line H2a side along the longitudinal direction of the second reference flat region 11b is defined as a fourth end joint portion 6h.

[0156] An imaginary line passing through an end surface **14***c* of the fourth shortest joint portion **6***g* on the fourth imaginary line H**2***a* side and parallel to the sheet thickness direction of the second reference flat region **11***b* is defined as an imaginary line G. An imaginary line passing through an end surface **14***d* of the fourth end joint portion **6***h* on the fourth imaginary line H**2***a* side and parallel to the sheet thickness direction of the second reference flat region **11***b* is defined as an imaginary line H. Among the joint portions **6** of the flat part **4***d* having the second reference flat region **11***b*, a joint portion **6** located between the imaginary line G and the imaginary line H is defined as the fourth-group joint portion V.sub.2O. Here, the number of fourth-group joint portions V.sub.2O is m (m is a natural number) in total from V.sub.2Ol to V.sub.2Om.
(Average Distance of Fourth-Group Joint Portions <L.SUB.2O.>)
[0157] An average of lengths from the fourth imaginary line H**2***a* to the end surface **14** of the fourth-group joint portions V.sub.2O on the fourth imaginary line H**2***a* side along the Longitudinal direction of the second reference flat region **11***b* is defined as an average distance of fourth-group joint portions V.sub.2o <L.sub.2O>. The average distance of fourth-group joint portions V.sub.2o <L.sub.O> can be measured by the following method. An observation image of the side surface of the wound core is obtained using an optical microscope or the like. In the obtained observation image, the fourth-group joint portion V.sub.2o is specified based on the definition described above. Next, length L.sub.2O from the fourth imaginary line H**2***a* to the end surface of the fourth-group joint portion V.sub.2O on the fourth imaginary line H**2***a* side along the longitudinal direction of the second reference flat region **11***b* is measured using image processing software. An average value of the obtained L.sub.2O is obtained, and the average value is defined as the average distance of fourth-group joint portions <L.sub.2O>.
[0158] In the wound core **10**F, the joint portions **6** are preferably arranged such that the joint portions **6** are shifted from each other in a stepwise manner in the circumferential direction. The circumferential position of the joint portion **6** in the bent body **1** is gradually shifted from the third imaginary line H**1***a* side (first-group joint portion Vi side) to the fourth imaginary line H**2***a* side (second-group joint portion Vo side) in the circumferential direction from the bent body **1** located on the inner side in the radial direction toward the bent body **1** located on the outer side in the radial direction. In the flat part **4***d*, the joint portions **6** are arranged such that a plurality of stepwise patterns are repeated in the radial direction. In the wound core **10**F, among the joint portions **6** arranged in one stepwise pattern, a joint portion **6** of the bent body **1** located on the innermost side in the radial direction is included in the third-group joint portion V.sub.2i, and a joint portion **6** of the bent body **1** located on the outermost side in the radial direction is included in the fourth-group joint portion V.sub.2o. By sequentially shifting the joint portions **6** along the circumferential direction in this manner, it is possible to suppress inhibition of a flow of magnetic flux in the wound core **10**F.
[0159] In the wound core **10**F, the number of third-group joint portions V.sub.2i is preferably equal to the number of fourth-group joint portions V.sub.2o. Also, in the wound core **10**F, among a second quotient and a second remainder obtained by dividing the number of joint portions **6** in the flat part **4***d* located between the third imaginary line H**1***a* and the fourth imaginary line H**2***a* and having the second reference flat region **11***b* by the number of third-group joint portions V.sub.2i, the second quotient is defined as k2, and k2 satisfies the following expression (6). In FIG. **11**, the number k2 is equal to the number of joint portions located between V.sub.2i1 and V.sub.2o1 and located between the third imaginary line H**1***a* and the fourth imaginary line H**2***a* along the sheet thickness direction. That is, k2 is the number of joint portions **6** arranged to be shifted stepwise from the specific third-group joint portion V.sub.2i to the fourth-group joint portion V.sub.2o closest to the third-group joint portion V.sub.2i. The number k2 is the number of joint portions included in one stepwise pattern. By arranging the joint portion **6** in this manner, noise can be further suppressed.

[00009] $9 \leq k2 \leq 20$ (6)

<Wound Core Manufacturing Method>

[0160] Next, the wound core manufacturing method of the present disclosure will be described. Also, a method of manufacturing the grain-oriented electrical steel sheet constituting the bent body **1** is not particularly limited, and a method of manufacturing a conventionally known grain-oriented electrical steel sheet can be appropriately selected. Preferred specific examples of the manufacturing method include a method in which a slab having a chemical composition of the grain-oriented electrical steel sheet is heated to 1000° C. or higher to perform hot rolling, and then hot-band annealing is performed as necessary, then cold rolling is performed once or twice or more with intermediate annealing interposed therebetween to obtain a cold-rolled steel sheet, and the cold-rolled steel sheet is heated to 700 to 900° C. in, for example, a wet hydrogen-inert gas atmosphere to perform decarburization annealing, nitriding annealing is further performed as necessary, an annealing separator is applied, then final annealing is performed at about 1000° C., and thus an insulating coating is formed at about 900° C. Thereafter, coating or the like may be further performed for adjusting the dynamic friction coefficient.

[0161] In the wound core manufacturing method of the present disclosure, the wound core **10** including the grain-oriented electrical steel sheets each having the above-described form is manufactured by shearing, folding, and laminating the grain-oriented electrical steel sheets in the sheet thickness direction such that the average distance of first-group joint portions V.sub.i<L.sub.i> satisfies the above expression (1) and the average distance of second-group joint portions V.sub.O<L.sub.O> satisfies the above expression (2) when the bent body **1** has one joint portion **6**. Also, when there are two joint portions **6** in the bent body **1**, it is preferable that the grain-oriented electrical steel sheets are sheared, folded, and laminated in the sheet thickness direction such that the average distance of first-group joint portions V.sub.i<L.sub.i> satisfies the above expression (1), the average distance of second-group joint portions V.sub.O<L.sub.O> satisfies the above expression (2), and the average distance of third-group joint portions V.sub.2i <L.sub.2i> and the average distance of fourth-group joint portions V.sub.2o <L.sub.2O> satisfy the above expressions (4) and (5). Each winding is assembled such that the end surfaces of the grain-oriented electrical steel sheets face each other via at least one joint portion **6**. The manufacturing method of the present disclosure manufactures a wound core satisfying the above conditions by adjusting a feed amount of the grain-oriented electrical steel sheet, a bending timing, and a shearing timing of the grain-oriented electrical steel sheet.

(Wound Core Manufacturing Apparatus)

[0162] Next, a wound core manufacturing apparatus according to the present disclosure will be described. The following manufacturing apparatus is an example of a manufacturing apparatus for manufacturing the wound core **10** of the present disclosure. As illustrated in FIG. **12**, a wound core manufacturing apparatus **40** is a manufacturing apparatus **40** of the wound core **10** formed by bending and laminating steel sheets (grain-oriented electrical steel sheets) **21**. The wound core manufacturing apparatus **40** includes a bending device **20** that bends the grain-oriented electrical steel sheet **21** and a feed roll **60** that feeds the grain-oriented electrical steel sheet **21** to the bending device **20**. The wound core manufacturing apparatus **40** of the present disclosure may include a decoiler **50** and a cutting device **70**.

“Decoiler”

[0163] The decoiler **50** unwinds the grain-oriented electrical steel sheet **21** from a coil **27** of the grain-oriented electrical steel sheet **21**. The grain-oriented electrical steel sheet **21** unwound from the decoiler **50** is conveyed toward the feed roll **60**.

“Feed Roll”

[0164] The feed roll **60** conveys the grain-oriented electrical steel sheet **21** to the bending device **20**. The feed roll **60** adjusts a conveyance direction **25** of the grain-oriented electrical steel sheet **21** immediately before being supplied into the bending device **20**. The feed roll **60** adjusts the conveyance direction **25** of the grain-oriented electrical steel sheet **21** in a horizontal direction, and then supplies the grain-oriented electrical steel sheet **21** to the bending device **20**.

[0165] The cutting device **70** is installed between the feed roll **60** and the bending device **20**. The grain-oriented electrical steel sheet **21** is cut by the cutting device **70**, and then bent. The cutting method is not particularly limited. The cutting method is, for example, shearing.

“Bending Device”

[0166] The bending device **20** bends the grain-oriented electrical steel sheet **21** conveyed from the feed roll **30**. A bent body **1** has a bent region obtained by bending and a flat region adjacent to the bent region. In the bent body **1**, a bent body flat part and a bent body corner portion are alternately continuous. In each corner portion, an angle formed by two adjacent flat parts is preferably substantially 90°.

[0167] The bending device **20** includes, for example, a die **22** and a punch **24** for press working. The bending device further includes a guide **23** for fixing the grain-oriented electrical steel sheet **21** and a cover (not illustrated). The cover covers the die **22**, the punch **24**, and the guide **23**. After the bending device **20** bends the grain-oriented electrical steel sheet **21**, the grain-oriented electrical steel sheet **21** may be cut by the cutting device **70**. After the cutting device **70** cuts the grain-oriented electrical steel sheet **21**, the bending device **20** may perform bending.

[0168] The grain-oriented electrical steel sheet **21** is conveyed in the conveyance direction **25** and fixed at a position set in advance. Next, the punch **24** pressurizes up to a predetermined position in a pressurization direction **26** with a predetermined force set in advance, so that the bent body **1** having a bent region of a desired bending angle φ is obtained.

“Lamination”

[0169] By the bending device **20**, the bent bodies **1** are laminated in a sheet thickness direction. The bent bodies **1** are laminated by aligning bent body corner portions **3** and being overlapped in a sheet thickness direction to form, for example, a laminated body **2** having a substantially rectangular shape in viewing from the side. As a result, it is possible to obtain the low-noise wound core according to the present disclosure. When the number of joint portions **6** of the bent body **1** is one, the bent bodies **1** are laminated in the sheet thickness direction by the bending device **20** such that the average distance of first-group joint portions V.sub.i<L.sub.i> and the average distance of second-group joint portions V.sub.O<L.sub.O> satisfy the above expressions (1) and (2). When there are two joint portions **6** in the bent body **1**, it is preferable to laminate the bent bodies **1** in the sheet thickness direction such that the average distance of first-group joint portions V.sub.i<L.sub.i> and the average distance of second-group joint portions V.sub.O<L.sub.O> satisfy the above expressions (1) and (2), and the average distance of third-group joint portions V.sub.2i <L.sub.2i> and the average distance of fourth-group joint portions V.sub.2o<L.sub.2O> satisfy the above expressions (4) and (5). The obtained wound core may be further fixed using a known binding band or fastening tool as necessary.

[0170] The present disclosure is not limited to the above embodiments. The above embodiments are examples, and anything having substantially the identical configuration as the technical idea described in the claims of the present disclosure and exhibiting the same operation and effects is included in the technical scope of the present disclosure. The wound core manufacturing method of the present disclosure manufactures a wound core using the above wound core manufacturing apparatus.

EXAMPLES

[0171] Hereinafter, examples (experimental examples) will be described, but the wound core according to the present disclosure is not limited to the following examples. The wound core of the present disclosure can adopt various conditions as long as the object of the present disclosure is achieved without departing from the gist of the present disclosure. The conditions in the following examples are condition examples adopted to confirm the operability and effects.

Experimental Example 1

[Manufacture of Wound Core]

[0172] Grain-oriented electrical steel sheets having sheet thicknesses in Tables 1A to 1J (sheet width: 152.4 mm, sheet thickness: 0.23 mm or 0.18 mm, Si content: 3.45 mass %) were sheared and bent so that the average distance of first-group joint portions V.sub.i<L.sub.i>, the average distance of second-group joint portions V.sub.O<L.sub.O>, the average distance of third-group joint portions V.sub.2i <L.sub.2i>, the average distance of fourth-group joint portions V.sub.2o<L.sub.2O>, the number k, and the number k2 in Tables 2A to 2J were obtained to prepare bent bodies, and the bent bodies were laminated in the sheet thickness direction to obtain a wound core having dimensions shown in FIG. **13**. The bending angle φ of the wound core was set to 45°. L1 is a length of a flat part parallel to the X-axis direction. L2 is a length of a flat part parallel to the Z-axis direction. L3 is a winding thickness (thickness in the laminating direction) of the wound core. L4 is a circumferential length of a flat region of the innermost circumference at the corner portion of the wound core. In each example, L1: 344 mm, L2: 122 mm, L3: 94.1 mm, and LA: 4 mm were set. Also, the radius of curvature in each bent region was set to 1.5 mm. Although the joint portions are omitted in FIG. **13**, the joint portions of each example were formed in the above-described stepwise pattern. A wound core having one joint portion was defined as a core A, and a wound core having two joint portions was defined as a core B. Two joint portions of each bent body of the core B are in two flat regions facing each other. In Tables 2A to 2J, the column of joint portion **1** means a joint portion of a flat part having a reference flat region, and a joint portion **2** means a joint portion of a flat part having a second reference flat region. When each of the two flat parts had a joint portion, and only a plurality of joint portions of one flat part satisfied the conditions of the average distance of the above expressions (1) and (2), the joint portion of the flat part satisfying the conditions of the average distance of the above expressions (1) and (2) was defined as the joint portion **1**.

[Evaluation of Noise]

[0173] In noise measurement, the wound cores of Experiment Nos. 1 to 238 in Tables 1A to 1J were prepared and excited, and the noise measurement was performed. This noise measurement was performed in an anechoic chamber with a background noise of 16 dBA with a noise meter installed at a position of 0.3 m from the core surface using an A-weighted network. In the excitation, the frequency was set to 50 Hz, and the magnetic flux density was set to 1.7 T. A core noise of 45 dBA or less was regarded as acceptable.

TABLE-US-00001 TABLE 1A Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 1 A 0.23 94.1 418 2 A 0.23 94.1 418 3 A 0.23 94.1 418 4 A 0.23 94.1 418 5 A 0.23 94.1 418 6 A 0.23 94.1 418 7 A 0.23 94.1 418 8 A 0.23 94.1 418 9 A 0.23 94.1 418 10 A 0.23 94.1 418 11 A 0.23 94.1 418 12 A 0.23 94.1 418 13 A 0.23 94.1 418 14 A 0.23 94.1 418 15 A 0.23 94.1 418 16 A 0.23 94.1 418 17 A 0.23 94.1 418 18 A 0.23 94.1 418 19 A 0.23 94.1 418 20 A 0.23 94.1 418 21 A 0.23 94.1 418 22 A 0.23 94.1 418 23 A 0.23 94.1 418 24 A 0.23 94.1 418 25 A 0.23 94.1 418

TABLE-US-00002 TABLE 1B Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 26 A 0.23 94.1 418 27 A 0.23 94.1 418 28 A 0.23 94.1 418 29 A 0.23 94.1 418 30 A 0.23 94.1 418 31 A 0.23 94.1 418 32 A 0.23 94.1 418 33 A 0.23 94.1 418 34 A 0.23 94.1 418 35 A 0.23 94.1 418 36 A 0.23 94.1 418 37 A 0.23 94.1 418 38 A 0.23 94.1 418 39 A 0.23 94.1 418 40 A 0.23 94.1 418 41 A 0.23 94.1 418 42 A 0.23 94.1 418 43 A 0.23 94.1 418 44 A 0.23 94.1 418 45 A 0.23 94.1 418 46 A 0.23 94.1 418 47 A 0.23 94.1 418 48 A 0.23 94.1 415 49 A 0.23 94.1 416 50 A 0.23 94.1 417

TABLE-US-00003 TABLE 1C Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 51 A 0.23 94.1 419 52 A 0.23 94.1 420 53 A 0.23 94.1 414 54 A 0.23 94.1 418 55 A 0.23 94.1 418 56 A 0.23 94.1 418 57 A 0.23 94.1 418 58 A 0.23 94.1 418 59 A 0.23 94.1 418 60 A 0.23 94.1 418 61 A 0.23 94.1 418 62 A 0.23 94.1 418 63

A 0.23 94.1 418 64 A 0.23 94.1 418 65 A 0.23 94.1 414 66 A 0.23 94.1 415 67 A 0.23 94.1 417 68 A 0.23 94.1 419 69 A 0.23 94.1 420 70 A 0.23 94.1 422 71 A 0.23 94.1 418 72 A 0.23 94.1 418 73 A 0.23 94.1 418 74 A 0.23 94.1 418 75 A 0.23 94.1 418

TABLE-US-00004 TABLE 1D Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 76 A 0.23 94.1 418 77 A 0.23 94.1 418 78 A 0.23 94.1 418 79 A 0.23 94.1 418 80 A 0.23 94.1 418 81 A 0.23 94.1 422 82 A 0.23 94.1 416 83 A 0.23 94.1 418 84 A 0.23 94.1 418 85 A 0.23 94.1 418 86 A 0.23 94.1 418 87 A 0.23 94.1 418 88 A 0.23 94.1 418 89 A 0.23 94.1 418 90 A 0.23 94.1 418 91 A 0.23 94.1 418 92 A 0.23 94.1 418 93 A 0.23 94.1 418 94 A 0.23 94.1 418 95 A 0.23 94.1 418 96 A 0.23 94.1 418 97 A 0.23 94.1 418 98 A 0.23 94.1 418 99 A 0.23 94.1 418 100 A 0.23 94.1 418

TABLE-US-00005 TABLE 1E Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 101 A 0.23 94.1 418 102 A 0.23 94.1 418 103 A 0.23 94.1 418 104 A 0.23 94.1 418 105 A 0.23 94.1 418 106 A 0.23 94.1 418 107 A 0.23 94.1 418 108 A 0.23 94.1 418 109 A 0.23 94.1 418 110 A 0.23 94.1 418 111 A 0.23 94.1 418 112 A 0.23 94.1 418 113 A 0.23 94.1 418 114 A 0.23 94.1 418 115 A 0.23 94.1 418 116 A 0.23 94.1 418 117 A 0.23 94.1 418 118 A 0.23 94.1 418 119 A 0.23 94.1 418 120 A 0.23 94.1 418 121 A 0.23 94.1 418 122 A 0.23 94.1 418 123 A 0.23 94.1 418 124 A 0.23 94.1 418 125 A 0.23 94.1 418

TABLE-US-00006 TABLE 1F Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 126 A 0.23 94.1 418 127 A 0.23 94.1 418 128 A 0.23 94.1 418 129 A 0.23 94.1 418 130 A 0.23 94.1 418 131 A 0.23 94.1 418 132 A 0.23 94.1 418 133 A 0.23 94.1 418 134 A 0.23 94.1 418 135 A 0.23 94.1 418 136 A 0.23 94.1 418 137 A 0.23 94.1 418 138 A 0.23 94.1 418 139 A 0.23 94.1 418 140 A 0.23 94.1 418 141 A 0.23 94.1 418 142 A 0.23 94.1 418 143 A 0.23 94.1 418 144 A 0.23 94.1 418 145 A 0.23 94.1 418 146 A 0.23 94.1 418 147 A 0.23 94.1 418 148 A 0.23 94.1 418 149 A 0.23 94.1 418 150 A 0.23 94.1 418

TABLE-US-00007 TABLE 1G Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 151 A 0.23 94.1 418 152 A 0.23 94.1 418 153 A 0.23 94.1 418 154 A 0.23 94.1 418 155 A 0.23 94.1 418 156 A 0.23 94.1 418 157 A 0.23 94.1 418 158 A 0.23 94.1 418 159 A 0.23 94.1 418 160 A 0.23 94.1 418 161 A 0.23 94.1 418 162 A 0.23 94.1 418 163 A 0.23 94.1 418 164 A 0.23 94.1 418 165 A 0.23 94.1 418 166 A 0.23 94.1 418 167 A 0.23 94.1 418 168 A 0.23 94.1 418 169 A 0.23 94.1 418 170 A 0.23 94.1 418 171 A 0.23 94.1 418 172 A 0.23 94.1 418 173 A 0.23 94.1 418 174 A 0.23 94.1 418 175 A 0.23 94.1 418

TABLE-US-00008 TABLE 1H Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 176 A 0.23 94.1 418 177 A 0.23 94.1 418 178 A 0.23 94.1 418 179 A 0.23 94.1 418 180 A 0.23 94.1 418 181 A 0.23 94.1 418 182 A 0.23 94.1 418 183 A 0.23 94.1 418 184 A 0.23 94.1 418 185 A 0.23 94.1 418 186 A 0.23 94.1 418 187 A 0.23 94.1 418 188 A 0.23 94.1 418 189 A 0.23 94.1 418 190 A 0.23 94.1 418 191 A 0.23 94.1 418 192 A 0.23 94.1 418 193 A 0.23 94.1 418 194 A 0.23 94.1 418 195 B 0.23 94.1 418 196 B 0.23 94.1 418 197 B 0.23 94.1 418 198 B 0.23 94.1 422 199 B 0.23 94.1 425 200 B 0.23 94.1 417

TABLE-US-00009 TABLE 1I Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 201 B 0.23 94.1 418 202 B 0.23 94.1 418 203 B 0.23 94.1 418 204 B 0.23 94.1 418 205 B 0.23 94.1 418 206 B 0.23 94.1 418 207 A 0.18 94.1 418 208 A 0.18 94.1 418 209 A 0.18 94.1 418 210 A 0.18 94.1 418 211 A 0.18 94.1 418 212 A 0.18 94.1 418 213 A 0.18 94.1 418 214 A 0.18 94.1 418 215 A 0.18 94.1 418 216 A 0.18 94.1 418 217 A 0.18 94.1 531 218 B 0.18 94.1 531 219 B 0.18 94.1 531 220 B 0.18 94.1 531 221 B 0.18 94.1 531 222 B 0.18 94.1 534 223 B 0.18 94.1 527 224 B 0.18 94.1 531 225 B 0.18 94.1 531

TABLE-US-00010 TABLE 1J Number of Material sheet Winding laminated Experiment Core thickness thickness sheets No. type (mm) (mm) (sheets) 226 B 0.18 94.1 531 227 B 0.18 94.1 531 228 B 0.18 94.1 531 229 B 0.23 94.1 418 230 B 0.23 94.1 418 231 B 0.23 94.1 418 232 B 0.23 94.1 418 233 B 0.23 94.1 418 234 B 0.18 94.1 531 235 B 0.18 94.1 531 236 B 0.18 94.1 531 237 B 0.18 94.1 531 238 B 0.18 94.1 531

TABLE-US-00011 TABLE 2A Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 1 1 1 8 56 2 2 2 8 56 3 5 5 8 56 4 9 9 8 56 5 15 15 8 56 6 20 20 8 56 7 25 25 8 56 8 30 30 8 56 9 40 40 8 56 10 1 2 8 56 11 2 4 8 45 12 5 10 8 45 13 9 18 8 45 14 15 30 8 45 15 20 40 8 45 16 26 50 8 56 17 30 55 8 56 18 40 65 8 56 19 1 3 8 56 20 2 6 8 37 21 5 15 8 37 22 9 27 8 37 23 15 45 8 37 24 20 60 8 37 25 26 65 8 46

TABLE-US-00012 TABLE 2B Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 26 30 70 8 56 27 40 75 8 56 28 1 10 8 56 29 2 20 8 42 30 5 50 8 42 31 9 90 8 42 32 15 150 8 42 33 20 200 8 42 34 26 250 8 56 35 30 300 8 56 36 40 300 8 56 37 1 1 9 56 38 2 2 9 56 39 5 5 9 56 40 9 9 9 56 41 15 15 9 56 42 20 20 9 56 43 26 26 9 56 44 30 30 9 56 45 40 40 9 56 46 1 2 9 56 47 2 4 9 38 48 2 4 9 38 49 2 4 9 38 50 2 4 9 38

TABLE-US-00013 TABLE 2C Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 51 2 4 9 38 52 2 4 9 38 53 2 4 9 38 54 5 10 9 38 55 9 18 9 38 56 15 30 9 38 57 20 40 9 38 58 26 50 9 56 59 30 55 9 56 60 40 65 9 56 61 1 3 9 56 62 2 6 9 28 63 5 15 9 28 64 9 27 9 28 65 9 27 9 28 66 9 27 9 28 67 9 27 9 28 68 9 27 9 28 69 9 27 9 28 70 9 27 9 28 71 15 45 9 28 72 20 60 9 28 73 26 65 9 54 74 30 70 9 54 75 40 75 9 54

TABLE-US-00014 TABLE 2D Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 76 1 10 9 54 77 2 20 9 38 78 5 50 9 38 79 9 90 9 38 80 15 150 9 38 81 15 150 9 38 82 15 150 9 38 83 20 200 9 38 84 26 250 9 56 85 30 300 9 56 86 40 300 9 56 87 1 1 13 56 88 2 2 13 56 89 5 5 13 56 90 9 9 13 56 91 15 15 13 56 92 20 20 13 56 93 26 26 13 56 94 30 30 13 56 95 40 40 13 56 96 1 2 13 56 97 2 4 13 40 98 5 10 13 40 99 9 18 13 40 100 15 30 13 40

TABLE-US-00015 TABLE 2E Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 101 20 40 13 40 102 26 50 13 56 103 30 55 13 56 104 40 65 13 56 105 1 3 13 56 106 2 6 13 35 107 5 15 13 35 108 9 27 13 35 109 15 45 13 35 110 20 60 13 35 111 26 65 13 56 112 30 70 13 56 113 40 75 13 56 114 1 10 13 56 115 2 20 13 44 116 5 50 13 44 117 9 90 13 44 118 15 150 13 44 119 20 200 13 44 120 26 250 13 56 121 30 300 13 56 122 40 300 13 56 123 1 1 20 56 124 2 2 20 56 125 5 5 20 56

TABLE-US-00016 TABLE 2F Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 126 9 9 20 56 127 15 15 20 56 128 20 20 20 56 129 26 26 20 56 130 30 30 20 56 131 40 40 20 56 132 1 2 20 56 133 2 4 20 38 134 5 10 20 38 135 9 18 20 38 136 15 30 20 38 137 20 40 20 38 138 26 50 20 56 139 30 55 20 56 140 40 65 20 56 141 1 3 20 56 142 2 6 20 38 143 5 15 20 38 144 9 27 20 38 145 15 45 20 38 146 20 60 20 38 147 26 65 20 56 148 30 70 20 56 149 40 75 20 56 150 1 10 20 56

TABLE-US-00017 TABLE 2G Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 151 2 20 20 38 152 5 50 20 38 153 9 90 20 38 154 15 150 20 38 155 20 200 20 38 156 26 250 20 56 157 30 300 20 56 158 40 300 20 56 159 1 1 23 56 160 2 2 23 56 161 5 5 23 56 162 9 9 23 56 163 15 15 23 56 164 20 20 23 56 165 26 26 23 56 166 30 30 23 56 167 40 40 23 56 168 1 2 23 56 169 2 4 23 45 170 5 10 23 45 171 9 18 23 45 172 15 30 23 45 173 20 40 23 45 174 26 50 23 56 175 30 55 23 56

TABLE-US-00018 TABLE 2H Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 176 40 65 23 56 177 1 3 23 56 178 2 6 23 37 179 5 15 23 37 180 9 27 23 37 181 15 45 23 37 182 20 60 23 37 183 26 65 23 56 184 30 70 23 56 185 40 75 23 56 186 1 10 23 56 187 2 20 23 42 188 5 50 23 42 189 9 90 23 42 190 15 150 23 42 191 20 200 23 42 192 26 250 23 56 193 30 300 23 56 194 40 300 23 56 195 1 3 13 1 3 13 59 196 2 6 13 2 6 13 29 197 5 15 13 5 15 13 29 198 5 15 13 5 15 13 29 199 5 15 13 5 15 13 29 200 5 15 13 5 15 13 29

TABLE-US-00019 TABLE 2I Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o>

<L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 201 9 27 13 9 27 13 29 202 15 45 13 15 45 13 29 203 20 60 13 20 60 13 29 204 26 65 13 26 65 13 59 205 30 70 13 30 70 13 59 206 40 75 13 40 75 13 59 207 30 55 13 56 208 40 65 13 56 209 1 3 13 56 210 2 6 13 35 211 5 15 13 35 212 9 27 13 35 213 15 45 13 35 214 20 60 13 35 215 26 65 13 56 216 30 70 13 56 217 40 75 13 56 218 1 3 13 1 3 13 59 219 2 6 13 2 6 13 29 220 5 15 13 5 15 13 29 221 9 27 13 9 27 13 29 222 9 27 13 9 27 13 29 223 9 27 13 9 27 13 29 224 15 45 13 15 45 13 29 225 20 60 13 20 60 13 29

TABLE-US-00020 TABLE 2J Joint portion 1 Joint portion 2 Experiment <L.sub.i> <L.sub.o> <L.sub.2i> <L.sub.2o> Core noise No. (mm) (mm) k (mm) (mm) k2 (dBA) 226 26 65 13 26 65 13 59 227 30 70 13 30 70 13 59 228 40 75 13 40 75 13 59 229 2 3 13 1 3 13 45 230 20 35 13 10 10 13 43 231 20 45 13 30 70 13 45 232 24 70 13 30 70 13 43 233 24 75 13 40 75 13 43 234 15 20 13 5 15 13 43 235 20 35 13 10 10 13 45 236 20 45 13 30 70 13 40 237 26 65 13 24 65 13 43 238 24 70 13 30 70 13 43

[0174] As shown in Tables 2A to 2J, in the case of the core A having one joint portion, when <Li> and <Lo> satisfied the above expressions (1) and (2), the noise was improved. In addition, when <Li> and <Lo> satisfied the above expressions (1) and (2), and the number k was 9 to 20, the noise was further improved.

[0175] Further, as shown in Tables 2A to 2J, in the case of the core having two joint portions, when <Li> and <Lo> satisfied the above expressions (1) and (2), and <L.sub.2i> and <L.sub.2O> satisfied the above expressions (4) and (5), the noise was improved. When <L.sub.i>, <Lo>, <L.sub.2i>, and <L.sub.2O> satisfied the above expressions (1), (2), (3), and (4), and the numbers k and k2 were 9 to 20, the noise was further improved.

Experimental Example 2

[Manufacture of Wound Core]

[0176] Grain-oriented electrical steel sheets having a sheet thickness in Table 3 (sheet width: 152.4 mm, sheet thickness: 0.23 mm or 0.18 mm, Si content: 3.45 mass %) were sheared and bent so that the average distance of first-group joint portions V.sub.i<L.sub.i>, the average distance of second-group joint portions V.sub.O<L.sub.O>, the average distance of third-group joint portions V.sub.2i <L.sub.2i>, the average distance of fourth-group joint portions V.sub.2o <L.sub.2O>, the number k, and the number k2 in Table 4 were obtained to prepare bent bodies, and the bent bodies were laminated in the sheet thickness direction to obtain a wound core in FIG. **13**. The bending angle of the bent region, the radius of curvature of the bent region, (the length between the imaginary line A and the imaginary line C along the longitudinal direction)/(the length between the first imaginary line H**1** and the second imaginary line H**2** along the longitudinal direction), and each dimension of each experimental example were set as shown in Table 3. One joint portion was provided in Experiment Nos. 1B, 3B, 4B, and 7B to 13B, and two joint portions were provided in Experiment Nos. 2B, 5B, and 6B. Two joint portions of each of the bent bodies of Experiment Nos. 2B, 5B, and 6B are in two flat regions facing each other. In Table 4K, the column of joint portion **1** means a joint portion of a flat part having a reference flat region, and a joint portion **2** means a joint portion of a flat part having a second reference flat region.

[Evaluation of Noise]

[0177] In noise measurement, the wound cores of Experiment Nos. 1B to 13B in Table 4 were prepared and excited, and the noise measurement was performed. This noise measurement was performed in an anechoic chamber with a background noise of 16 dBA with a noise meter installed at a position of 0.3 m from the core surface using an A-weighted network. In the excitation, the frequency was set to 50 Hz, and the magnetic flux density was set to 1.7 T. A core noise of 45 dBA or less was regarded as acceptable.

[0178] As shown in Table 4, the noise was improved in Experiment Nos. 1B to 9B and 11B to 13B. Also, when the number k was 9 to 20, the noise was further improved. In Experiment No. 10B, the noise was not improved because the radius of curvature exceeded 5.0 mm.

TABLE-US-00021 TABLE 3 Ratio of length between imaginary line A and imaginary line C/(length Radius between first Material Number of Bending of imaginary line H1 sheet Winding laminated Core angle curvature L1 L2 L3 L4 and second thickness thickness sheets specification (°) (mm) (mm) (mm) (mm) (mm) imaginary line H2) (mm) (mm) (sheets) a 45 0.2 344 122 94.1 4 0.50 0.23 94.1 418 b 45 1.0 344 122 94.1 4 0.50 0.23 94.1 418 b′ 45 1.0 344 122 94.1 4 0.90 0.23 94.1 418 c 45 2.9 344 122 94.1 4 0.50 0.23 94.1 418 c′ 45 2.9 344 122 94.1 4 0.90 0.23 94.1 418 d 45 4.9 344 122 94.1 4 0.50 0.23 94.1 418 e 45 10.0 344 122 94.1 4 0.50 0.23 94.1 418 f 30 1.0 344 122 94.1 4 0.50 0.23 94.1 418 g 30 2.9 344 122 94.1 4 0.50 0.23 94.1 418 h 60 1.0 344 122 94.1 4 0.50 0.23 94.1 418

TABLE-US-00022 TABLE 4 Core Joint portion 1 Joint portion 2 Core Experiment specifi- <Li> <Lo> <L.sub.2i> <L.sub.2o> noise No. cation (mm) (mm) k (mm) (mm) k2 (dBA) 1B a 9 27 9 29 2B a 9 27 9 9 27 9 29 3B b 5 10 9 40 4B b′ 5 10 9 39 5B b 5 15 13 5 15 13 40 6B b′ 5 15 13 5 15 13 39 7B c 9 27 9 32 8B c′ 9 27 9 30 9B d 5 10 9 40 10B e 15 30 20 47 11B f 5 15 13 30 12B g 15 30 20 39 13B h 5 15 13 29

Field of Industrial Application

[0179] According to the present disclosure, noise of a wound core can be suppressed. Therefore, industrial applicability is large.

BRIEF DESCRIPTION OF THE REFERENCE SYMBOLS

[0180] **1** Bent body [0181] **2** Laminated body [0182] **3** Corner portion [0183] **4**, **4***a*, **4***b* Flat part [0184] **5**, **5***a*, **5***b* Bent region [0185] **6** Joint portion [0186] **8** Flat region [0187] **10** Wound core [0188] **20** Bending device [0189] **40** Manufacturing apparatus [0190] **21** Grain-oriented electrical steel sheet [0191] **22** Die [0192] **23** Guide [0193] **24** Punch [0194] **25** Conveyance direction [0195] **26** Pressurization direction

Claims

1. A wound core formed by laminating, in a sheet thickness direction, a plurality of bent bodies formed from a grain-oriented electrical steel sheet, wherein the wound core has a plurality of flat parts and a plurality of corner portions, the bent body has a plurality of flat regions and a plurality of bent regions adjacent to the flat regions, a radius of curvature of each of the bent regions is 5.0 mm or less, the bent body has one or more joint portions in which end surfaces of the grain-oriented electrical steel sheets in a longitudinal direction face each other, and when the bent body disposed on the innermost side is defined as a first bent body and a flat region where the joint portion of the first bent body is present is defined as a reference flat region, the joint portion of each of the plurality of bent bodies is located in the flat part having the reference flat region, and in a side view of the wound core, when one bent region adjacent to the reference flat region is defined as a first bent region, the other bent region adjacent to the reference flat region is defined as a second bent region, an imaginary line passing through an end point of the first bent region on the reference flat region side and parallel to the sheet thickness direction of the reference flat region is defined as a first imaginary line, an imaginary line passing through an end point of the second bent region on the reference flat region side and parallel to the sheet thickness direction of the reference flat region is defined as a second imaginary line, among the joint portions of the flat part having the reference flat region, the joint portion located between the first imaginary line and the second imaginary line and having the shortest length from the first imaginary line to the end surface of the joint portion on the first imaginary line side along the longitudinal direction of the reference flat region is defined as a first shortest joint portion, among the joint portions in the bent bodies adjacent in the sheet thickness direction to the bent body having the first shortest joint portion, the joint portion located between the first imaginary line and the second imaginary line and having a shorter length from the first imaginary line to the end surface of the joint portion on the first imaginary line side along the longitudinal direction of the reference flat region is defined as a first end joint portion, among the joint portions of the flat part having the reference flat region, the joint portion located between the first imaginary line and the second imaginary line and having the shortest length from the second imaginary line to the end surface of the joint portion on the second imaginary line side along the longitudinal direction of the reference flat region is defined as a second shortest joint portion, among the joint portions in the bent body adjacent in the sheet thickness direction to the bent body having the second shortest joint portion, the joint portion located between the first imaginary line and the second imaginary line and having a shorter length from the second imaginary line to the end surface of the joint portion on the second imaginary line side along the longitudinal direction of the reference flat region is defined as a second end joint portion, an imaginary line passing through the end surface of the first shortest joint portion on the first imaginary line side and parallel to the sheet thickness direction of the reference flat region is defined as an imaginary line A, an imaginary line passing through the end surface of the first end joint portion on the first imaginary line side and parallel to the sheet thickness direction of the reference flat region is defined as an imaginary line B, an imaginary line passing through the end surface of the second shortest joint portion on the second imaginary line side and parallel to the sheet thickness direction of the reference flat region is defined as an imaginary line C, an imaginary line passing through the end surface of the second end joint portion on the second imaginary line side and parallel to the sheet thickness direction of the reference flat region is defined as an imaginary line D, among the joint portions of the flat part having the reference flat region, the joint portion located between the imaginary line A and the imaginary line B is defined as a first-group joint portion, among the joint portions of the flat part having the reference flat region, the joint portion located between the imaginary line C and the imaginary line D is defined as a second-group joint portion, an average of lengths from the first imaginary line to the end surface of each of the first-group joint portions on the first imaginary line side along the longitudinal direction of the reference flat region is defined as <L.sub.i>, and an average of lengths from the second imaginary line to the end surface of each of the second-group joint portions on the second imaginary line side along the longitudinal direction of the reference flat region is defined as <L.sub.O>, the wound core satisfies the following expressions (1) and (2), $2\text{mm} \leq$ .Math. L_i .Math. $< 25\text{mm}$ (1)

$1.22*$ .Math. Li .Math. $\leq$ .Math. L_O .Math. . (2)

2. The wound core according to claim 1, wherein the number of first-group joint portions is equal to the number of second-group joint portions, and among a quotient and a remainder obtained by dividing the number of joint portions in the flat part located between the first imaginary line and the second imaginary line and having the reference flat region by the number of first-group joint portions, k, which is the quotient, satisfies the following expression (3), $9 \leq k \leq 20$. (3)

3. The wound core according to claim 1, wherein the bent bodies have the joint portion in each of two flat regions facing each other, the first bent body has the reference flat region and a second reference flat region facing the reference flat region, and the joint portion of each of the plurality of bent bodies is located in the flat part having the reference flat region and the flat part having the second reference flat region, and in a side view of the wound core, when one bent region adjacent to the second reference flat region is defined as a third bent region, the other bent region adjacent to the second reference flat region is defined as a fourth bent region, an imaginary line passing through an end point of the third bent region on the second reference flat region side and parallel to the sheet thickness direction of the second reference flat region is defined as a third imaginary line, an imaginary line passing through an end point of the fourth bent region on the second reference flat region side and parallel to the sheet thickness direction of the second reference flat region is defined as a fourth imaginary line, among the joint portions of the flat part having the second reference flat region, the joint portion located between the third imaginary line and the fourth imaginary line and having the shortest length from the third imaginary line to the end surface of the joint portion on the third imaginary line side along the longitudinal direction of the second reference flat region is defined as a third shortest joint portion, among the joint portions in the bent bodies adjacent in the sheet thickness direction to the bent body having the third shortest joint portion, the joint portion located between the third imaginary line and the fourth imaginary line and having a shorter length from the third imaginary line to the end surface of the joint portion on the third imaginary line side along the longitudinal direction of the second reference flat region is defined as a third end joint portion, among the joint portions of the flat part having the second reference flat region, the joint portion located between the third imaginary line and the fourth imaginary line and having the shortest length from the fourth imaginary line to the end surface of the joint portion on the fourth imaginary line side along the longitudinal direction of the second reference flat region is defined as a fourth shortest joint portion, among the joint portions in the bent body adjacent in the sheet thickness direction to the bent body having the fourth shortest joint portion, the joint portion located between the third imaginary line and the fourth imaginary line and having a shorter length from the fourth imaginary line to the end surface of the joint portion on the fourth imaginary line side along the longitudinal direction of the second reference flat region is defined as a fourth end joint portion, an imaginary line passing through the end surface of the third shortest joint portion on the third imaginary line side and parallel to the sheet thickness direction of the second reference flat region is defined as an imaginary line E, an imaginary line passing through the end surface of the third end joint portion on the third imaginary line side and parallel to the sheet thickness direction of the second reference flat region is defined as an imaginary line F, an imaginary line passing through the end surface of the fourth shortest joint portion on the fourth imaginary line side and parallel to the sheet thickness direction of the second reference flat region is defined as an imaginary line G, an imaginary line passing through the end surface of the fourth end joint portion on the fourth imaginary line side and parallel to the sheet thickness direction of the second reference flat region is defined as an imaginary line H, among the joint portions of the flat part having the second reference flat region, the joint portion located between the imaginary line E and the imaginary line F is defined as a third-group joint portion, among the joint portions of the flat part having the second reference flat region, the joint portion located between the imaginary line G and the imaginary line H is defined as a fourth-group joint portion, an average of lengths from the third imaginary line to the end surface of each of the third-group joint portions on the third imaginary line side along the longitudinal direction of the second reference flat region is defined as <L.sub.2i>, and an average of lengths from the fourth imaginary line to the end surface of each of the fourth-group joint portions on the fourth imaginary line side along the longitudinal direction of the second reference flat region is defined as <L.sub.2O>, the wound core satisfies the following expressions (4) and (5), $2\text{mm} \leq$ .Math. L_{2i} .Math. $< 25\text{mm}$ (4)

$1.22*$ .Math. L_{2i} .Math. $\leq$ .Math. L_{2O} .Math. . (5)

4. The wound core according to claim 3, wherein the number of third-group joint portions is equal to the number of fourth-group joint portions, and among a second quotient and a second remainder obtained by dividing the number of joint portions in the flat part located between the third imaginary line and the fourth imaginary line and having the second reference flat region by the number of third-group joint portions, k2, which is the second quotient, satisfies the following expression (6), $9 \leq k2 \leq 20$. (6)

5. The wound core according to claim 1, wherein a bending angle of the bent region is 30° to 60°.